26900533 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECSTA300 - APPLIED STATISTICS AND DATA MINING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. What do you mean by splicing of index number? [U] / [CO1]
2. What is meant by weighting in index numbers? Why is it necessary? [K] / [CO1]
3. Define Quantity index and Price index. [K] / [CO1]
4. Differentiate between direct and indirect methods of standardization of death rates. [U] / [CO2]
5. Explain the meaning of q_x. [U] / [CO2]
6. What is Sample Registration System? [U] / [CO2]
7. How can life tables be used to compare mortality between two different populations or countries? [U] / [CO2]
8. Define false positive and false negative. [K] / [CO3]
9. Give the formula for lift of a rule. [U] / [CO3]

10. How is data mining used in sentiment/social media analysis? [U] / [CO4]

[2x7 = 14]

Part B

Short essay type questions

Answer any **4** questions

Each question carries **6** marks

11. Calculate Marshall Edgeworth's price index from the following data: [A] / [CO1]

Commodity	Base year		current year	
	Price	Quantity	Price	Quantity
Sandwich	200	15	350	20
Burger	350	10	500	15
Cake	500	20	700	10
Ice cream	210	15	300	10

12. Define Factor reversal test. Examine whether Laspeyre's and Fisher's index number satisfies this test. [An] / [CO1]

13. The vital statistics directorate published fertility data for province 1 (developing agriculture economy) and province 2 (urbanized industrial economy) each with millions of reproductive age women : [E] / [CO2]

Age group	Province 1		Province 2	
	Female population	Live births	Female population	live births
15-19	**3,600,000**	**180,000**	**2,800,000**	**28,000**
20-24	**3,400,000**	**578,000**	**3,100,000**	**248,000**
25-29	**3,100,000**	**465,000**	**3,200,000**	**384,000**
30-34	**2,800,000**	**252,000**	**2,900,000**	**261,000**
35-39	**2,500,000**	**125,000**	**2,500,000**	**100,000**
40-44	**2,200,000**	**33,000**	**2,100,000**	**21,000**
45-49	**1,800,000**	**3,600**	**1,700,000**	**1,700**

Compare Total fertility rate of both provinces.

14. a) Discuss any two merits and demerits of Crude birth rate and General fertility rate. [A] / [CO2]

b) An administrative territory recorded the following demographic statistics over and annual period :

Age group	Male population	Female population	Live birth
0-14	140,000	132,000	-
15-19	46,000	44,000	1100

Age group	Male population	Female population	Live birth
20-24	42,000	41,000	5330
25-29	40,000	39,000	4680
30-34	38,000	37,000	2590
35-39	35,000	34,000	1020
40-44	31,000	30,000	210
45-49	27,000	26,000	26
50 and above	87,000	92,000	-

Calculate the Crude birth rate and General fertility rate.Discuss which district exhibit higher fertility perfomance.

15. Explain the major challenges faced in outlier detection. [U] / [CO3]

16. Discuss the techniques used in data preprocessing and highlight their significance. [U] / [CO3]

[6x4 = 24]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

17. Calculate price index and quantity index numbers for the year 2025 with 2020 as base using [E] / [CO1]
    (i) Laspeyre's method
    (ii) Paasche's method
    (iii) Fisher's method

Commodity	Quantity		Expenditure	
	2020	2025	2020	2025
Tea	100	152	500	945
Bread	80	100	320	555
Meat	60	72	150	360
Miscellaneous	30	33	360	297

18. Discuss the four components of time series. [An] / [CO3]

[12x1 = 12]